Beyond Time-to-Fill

Onboarding Readiness and HR-IT Handoffs

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299	43	3,947
started hires	unresolved	person-days

Recruitment reporting often stops at requisition closure. I wanted to examine the operational interval that follows: the time between a hire starting and the organisation holding sufficient evidence that access, equipment and orientation are complete. In this fictional Ontario/B.C. employer, the analytical problem is as much about the arrival and correction of evidence as it is about elapsed time.

Primary finding

Mean unready days within first 20 operating days



0.813 days lower; 95% synthetic interval -0.559 to 2.046

The early-handoff cohort has a lower synthetic mean, but the interval includes zero. I would use this result to scope a prospective HR-IT handoff pilot, not to report a proven reduction in onboarding time. The simulation deliberately imposes an intervention effect; this exercise demonstrates measurement and inference mechanics.

Evidence status: all operational data and financial assumptions are synthetic. The figures below are model/fixture results, not client outcomes.

From an HR decision to a defensible metric

The business owner is a TA operations lead working with the HRIS service owner and IT service management. Their decision is whether initiating prerequisite work earlier would improve readiness without creating excessive rework, access risk or source-team workload. Canadian labour-market evidence motivates careful use of available capacity; it does not identify the effect at this fictional employer.

01 / Population and grain
300 synthetic hires are generated. The 30 September extract includes 299 whose start date is on or before the cutoff. Output grain is one started hire; exported task-event grain is one revision row.

02 / Temporal reconstruction
Filter recorded_date <= as_of before applying ROW_NUMBER over event_id, ordered by recorded_date descending and row_id descending. The row ID resolves same-day ties; effective_date describes business occurrence.

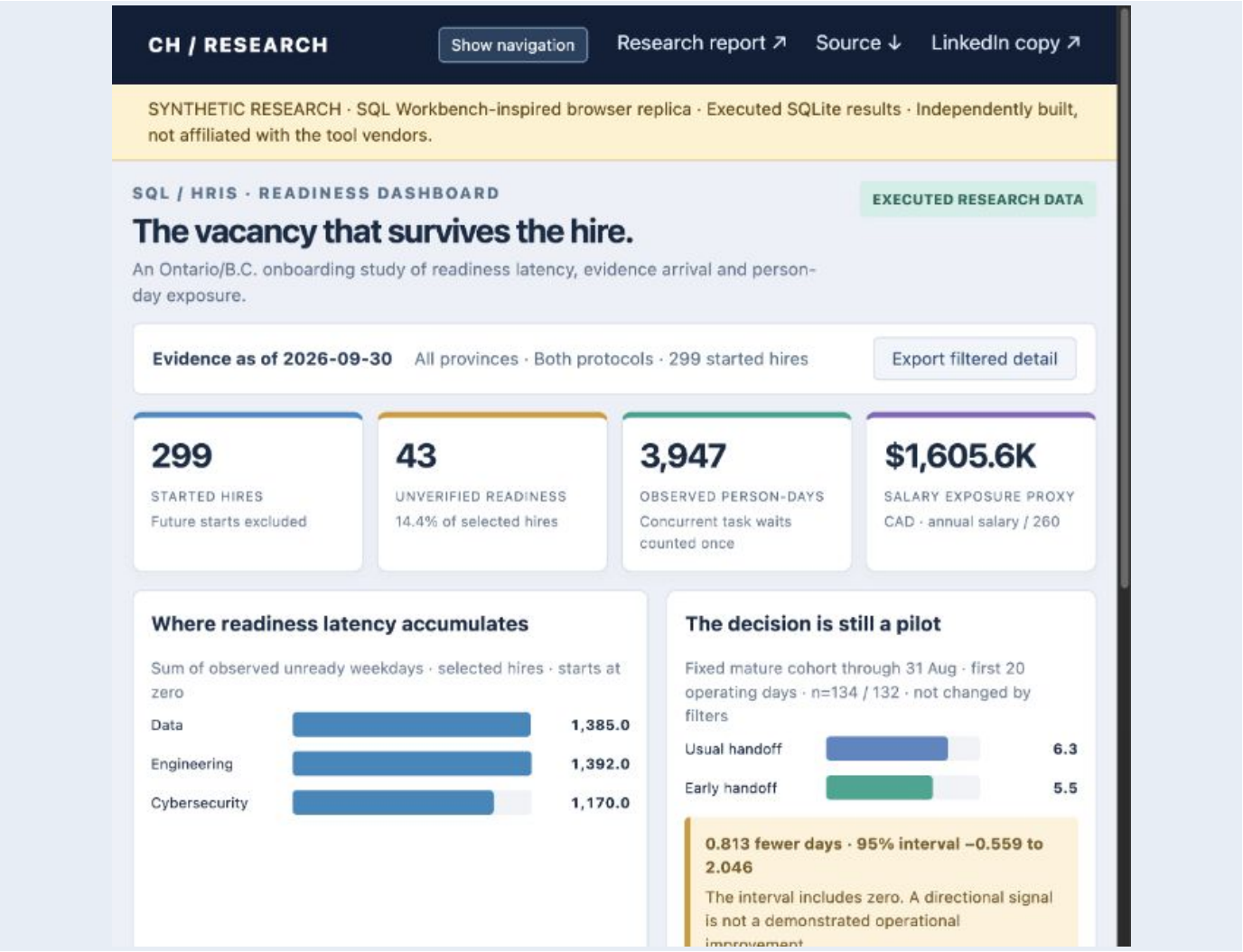
03 / Readiness endpoint
Three required task types must have qualifying completion evidence: access, equipment and orientation. Ready date is the later of start date and final required completion. Missing task streams remain unresolved through a LEFT JOIN.

04 / Exposure and comparison
Count calendar days that are operating days inside [start, readiness); include the cutoff for unresolved hires. For the cohort comparison, use starts through 31 August and restrict the outcome to the first 20 operating days.

Metric / construct	Definition / interpretation
Time to accept	Accepted date minus opened date; calendar days
Readiness latency	Unique unready operating days; one hire-day once
Right censoring	Readiness not verified by the analysis cutoff
Salary exposure	Unready weekdays × annual salary / 260
Contrast	Usual mean minus early-handoff mean; mature cohort

Reproducibility: the independent project package contains source, frozen synthetic inputs, a verification entry point, dashboard data and the report. Run **python3 verify.py** from the project root.

Management view: readiness is an evidence state



01 / Slicer context

The cutoff, province and protocol restrict the descriptive population. The fixed mature-cohort comparison is explicitly labelled and does not respond to those filters.

02 / Denominator discipline

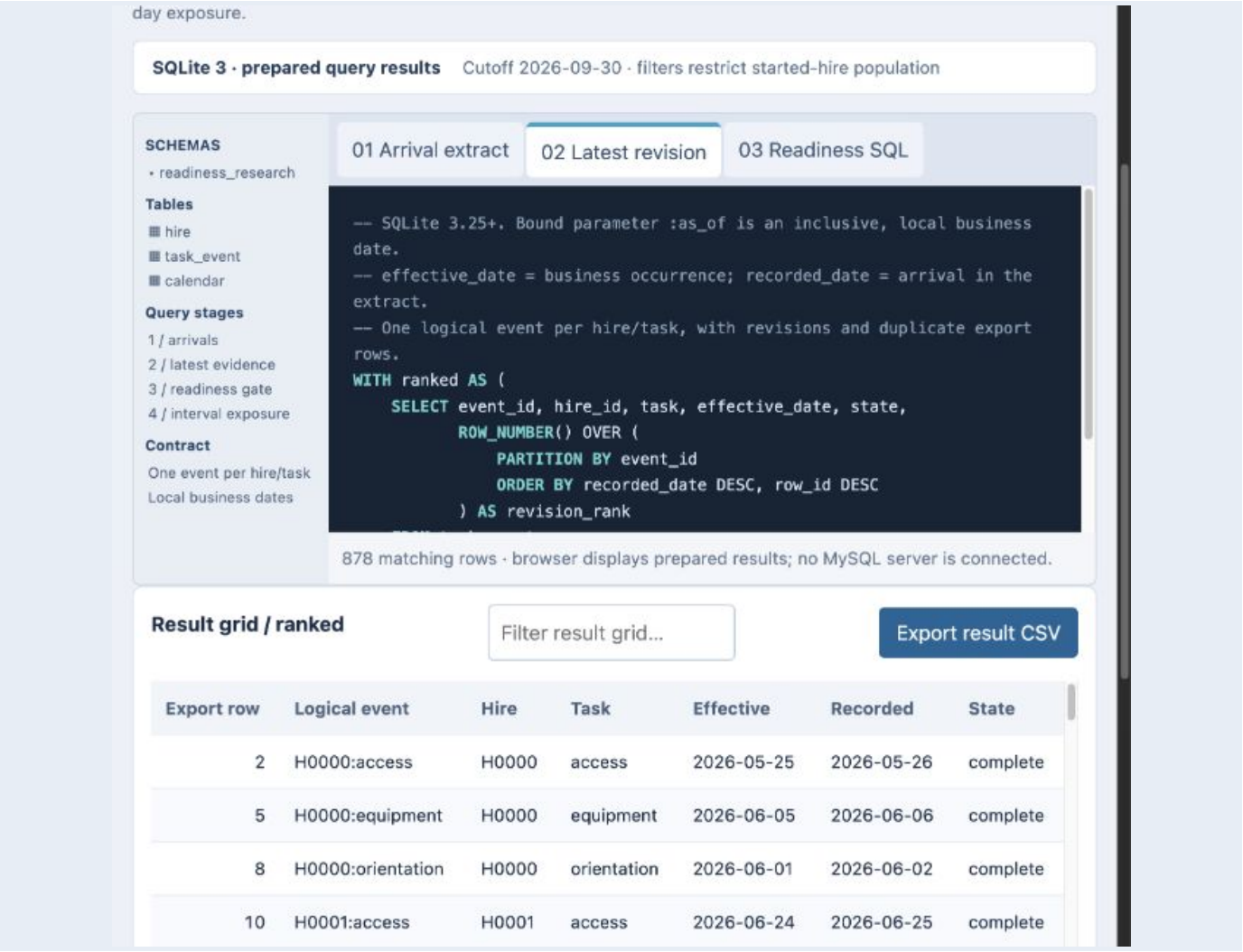
43 of 299 hires remain unresolved: 14.4%. A NULL ready date remains a state of evidence, not a zero-day completion.

03 / Exposure grain

The 3,947 weekdays count each hire-day once, even where several prerequisite tasks overlap.

Figure 1. Actual viewport capture of the working browser implementation. The scrollable dashboard contains the full charts, grids and source panes. Tool resemblance is a presentation convention; native execution is not implied. Full-size images and captions are in the screenshot gallery.

Technical view: reconstruct before aggregating



01 / Revision selection

The staged extract exposes arrival rows and the latest known revision separately. The result-grid filters preserve a trail back to each synthetic hire.

02 / SQL techniques

CTEs, ROW_NUMBER, conditional aggregation, LEFT JOIN and a calendar interval join implement the temporal contract.

03 / Tool provenance

This is a Workbench-inspired browser screen displaying executed SQLite results. The DrawSQL canvases are actual DrawSQL diagrams.

Figure 2. Actual viewport capture of the working browser implementation. The scrollable dashboard contains the full charts, grids and source panes. Tool resemblance is a presentation convention; native execution is not implied. Full-size images and captions are in the screenshot gallery.

What the evidence supports

Measure	Usual	Early handoff
Mature cohort n	134	132
Mean time to accept	29.64 days	29.31 days
Mean unready first 20 days	6.343 days	5.530 days

Validation and execution provenance

Evidence check	Result / boundary
CSV-to-database rebuild	299 unique output hires; 43 unresolved
Independent calendar oracle	All 299 person-day counts reconcile
Salary reconciliation	CAD 1,605,596.08 across selected hires
Teaching correction fixture	3, 5, 5 days at successive cutoffs
Historical integrity limit	Hire attributes are not historized

Limits of inference. The event-to-hire/task mapping remains an input assumption. The calendar is weekday-only and does not encode provincial holidays. Input date contracts and historical hire attributes need hardening. Salary exposure is not foregone profit or measured employee inactivity. The bootstrap describes the synthetic sample under its generator; it cannot establish external validity or causal effectiveness.

Recommended next study. Run a prospective pilot with an agreed service catalogue and completion-evidence standard. Assign a named HRIS data steward, record correction timestamps, add jurisdiction-specific operating calendars and monitor employee experience alongside readiness latency. Pre-register the endpoint, follow-up window and decision threshold before comparing protocols.

Sources and research basis

Statistics Canada: business conditions, Q2 2025

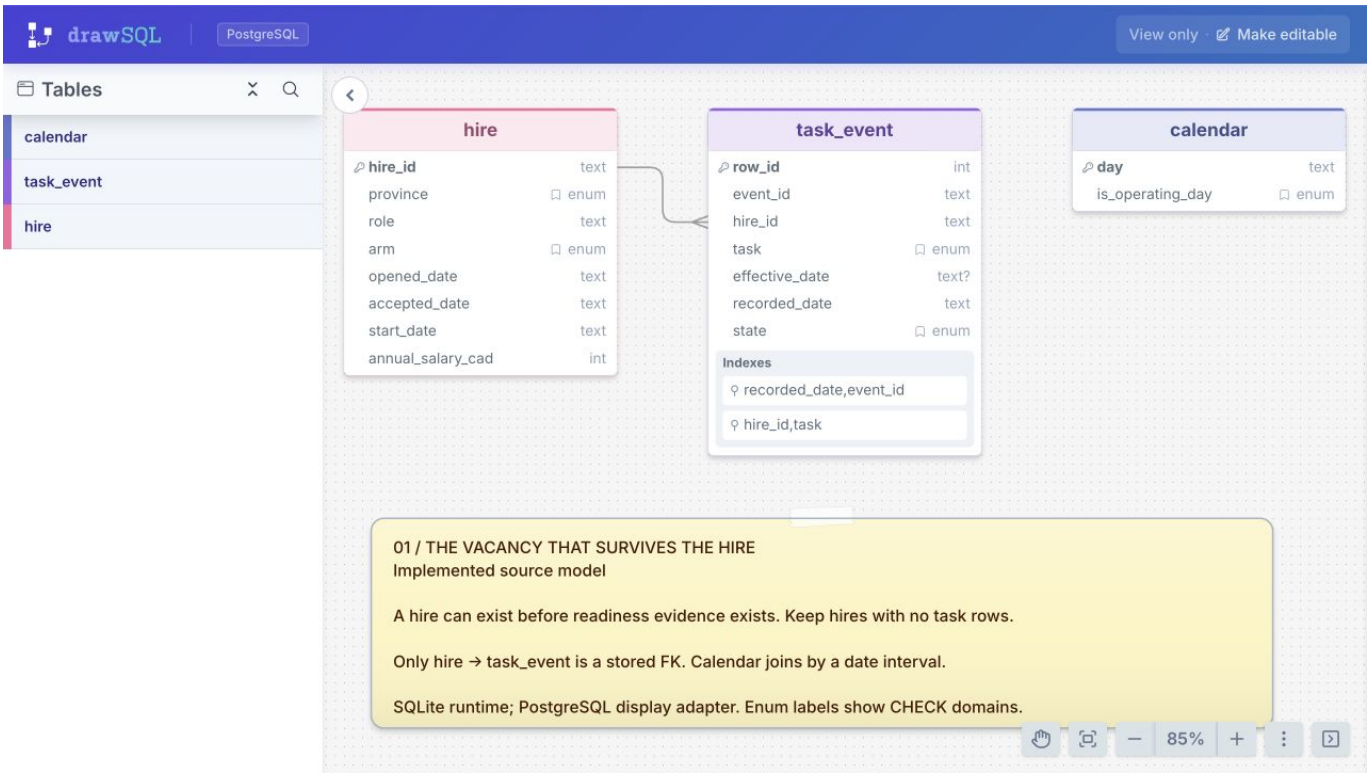
SQLite: window-function semantics

DrawSQL: account-free canvas and snapshot limits

Public sources establish context or method; they do not supply the simulated microdata. Ontario indexed official text was rechecked on 20 October 2026; the other sources were reviewed for the original research baseline. Full provenance is in SOURCES.md.

The evidence model behind the result

The implemented model keeps hire attributes, task-event revisions and operating dates separate. The hire-to-event relationship preserves the source grain. Two additional DrawSQL diagrams explain the evidence clock and the count-once exposure rule; those are conceptual teaching models, not additional database tables.



Actual DrawSQL browser capture. PostgreSQL import DDL adapts the implemented SQLite schema for visualization. Native export was not used.

DEMO-01: what changes when evidence arrives?

Cutoff	Ready date known	Unique weekdays	Censored
17 September	17 September	3	No
18 September	Unresolved	5	Yes
21 September	19 September	5	No

The 18 September revision retracts equipment completion. Later evidence restores completion effective on Saturday, 19 September. The reporting clock determines which revision is known; the operating calendar determines exposure. At the first cutoff, adding task waits would give five days even though only three distinct hire-days elapsed. This is a separate teaching fixture, not another study participant.

The interactive SQL showcase links all three DrawSQL diagrams and their import SQL. Local images preserve the presentation if a public snapshot expires. The fixture SQL and expected results are included for reproduction.